

Architecture of Expert System

joko purwadi

Materials

- Component of ES
- Architecture of ES

Basic Components

- ES has been structured in many ways
 - The various ES architecture include different components
- Basic components:
 1. User interface
 2. Knowledge base
 3. Inference engine

Basic Components (2)

1. User interface:

- A software that provides for the communication exchange between user and the system
- User: input facts, ask the system
- System: ask new facts, give answer or advice

2. Knowledge base:

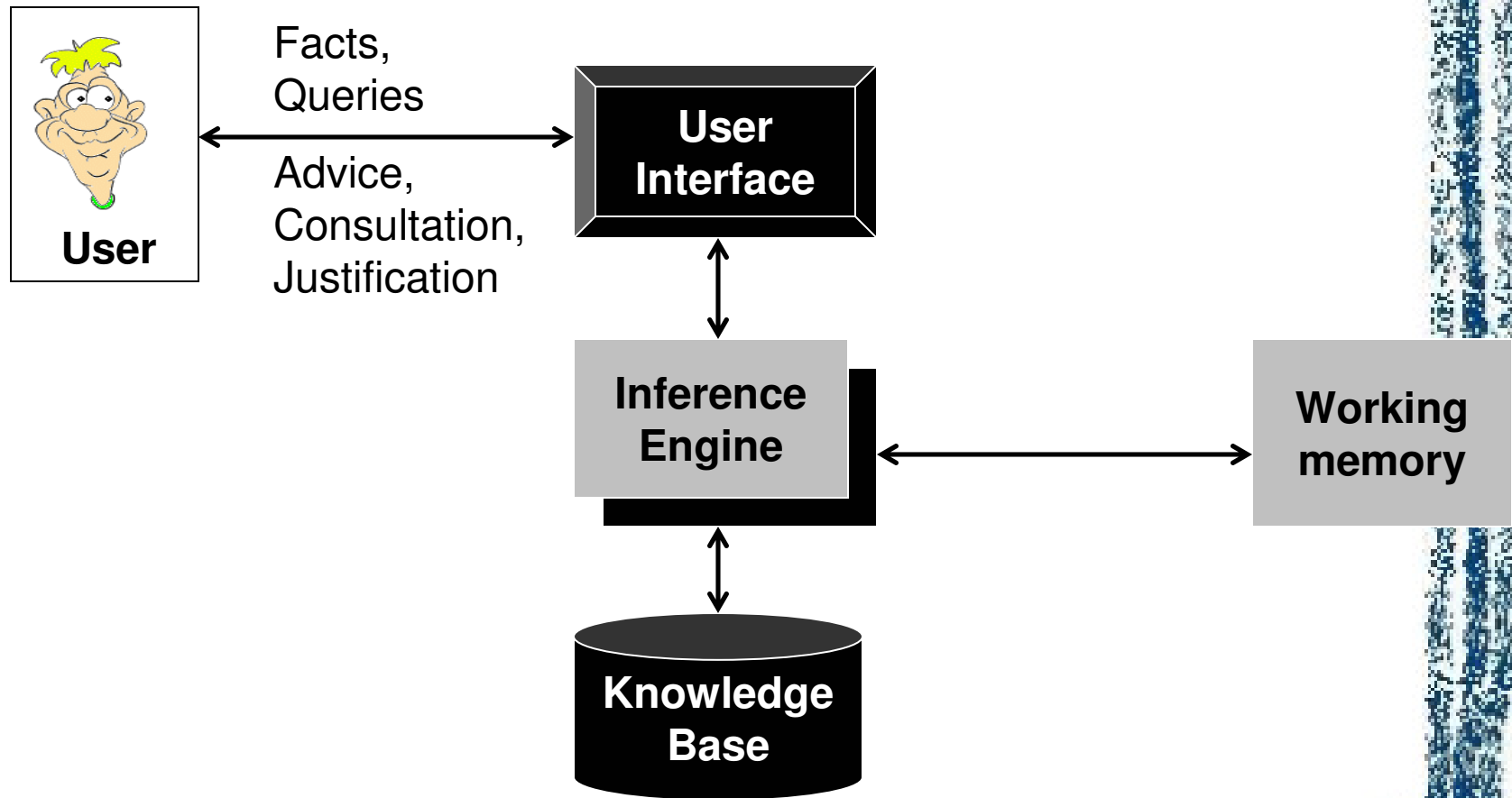
- Contains expert-level knowledge on a particular subject
- Stored in a knowledge representational form

3. Inference engine:

- A software that performs the inference reasoning tasks
- It uses the knowledge in the knowledge base and information provided by the user to infer new knowledge

Architecture of Expert System

A simple architecture

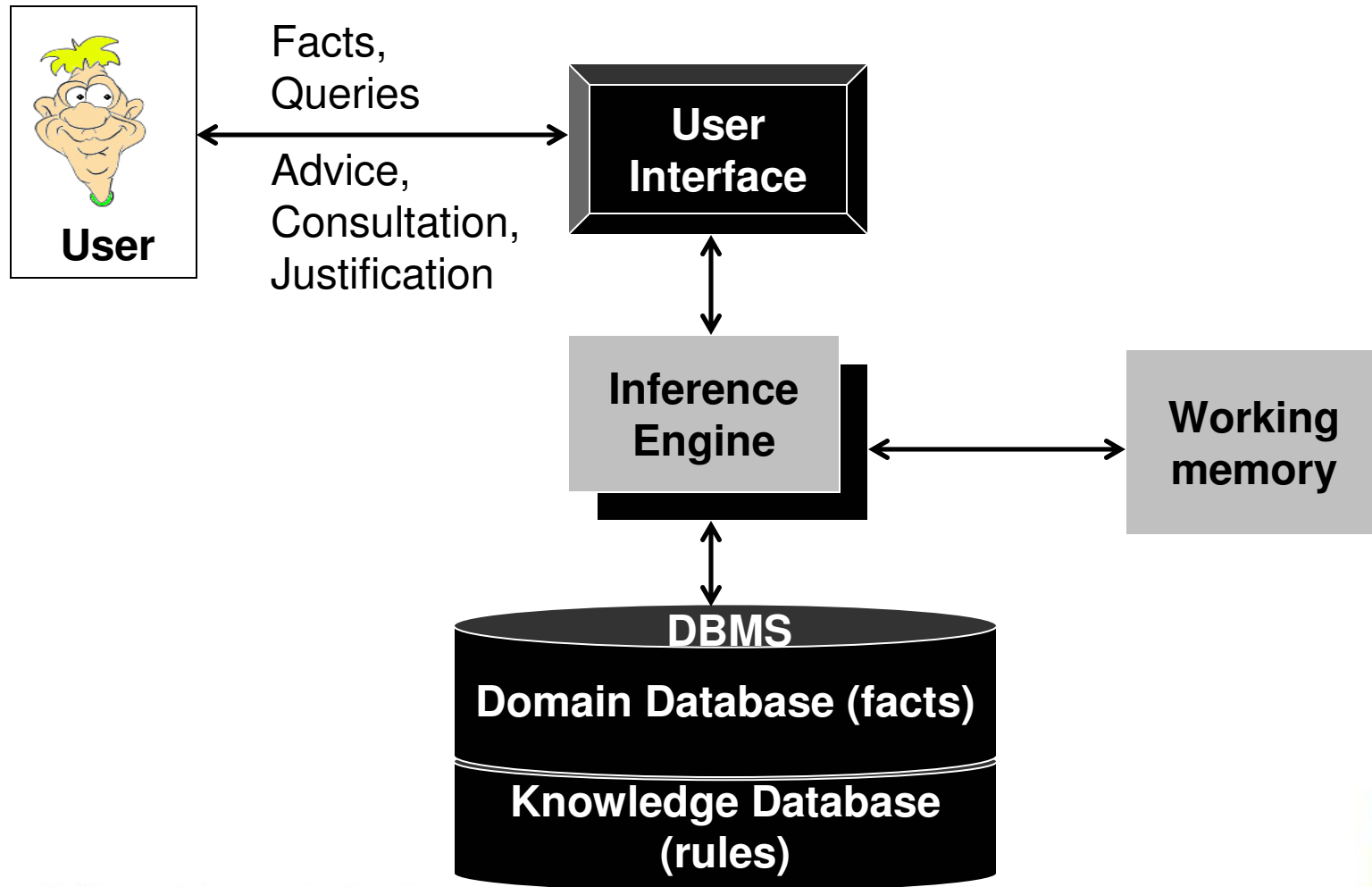


The Other Components

1. Working memory:
 - A global database of facts used by the rules
2. Knowledge database:
 - Contains rules about the behavior of the elements of a particular subject
3. Domain database:
 - Contains facts about the ES's subject

Architecture of Expert System

An extended architecture

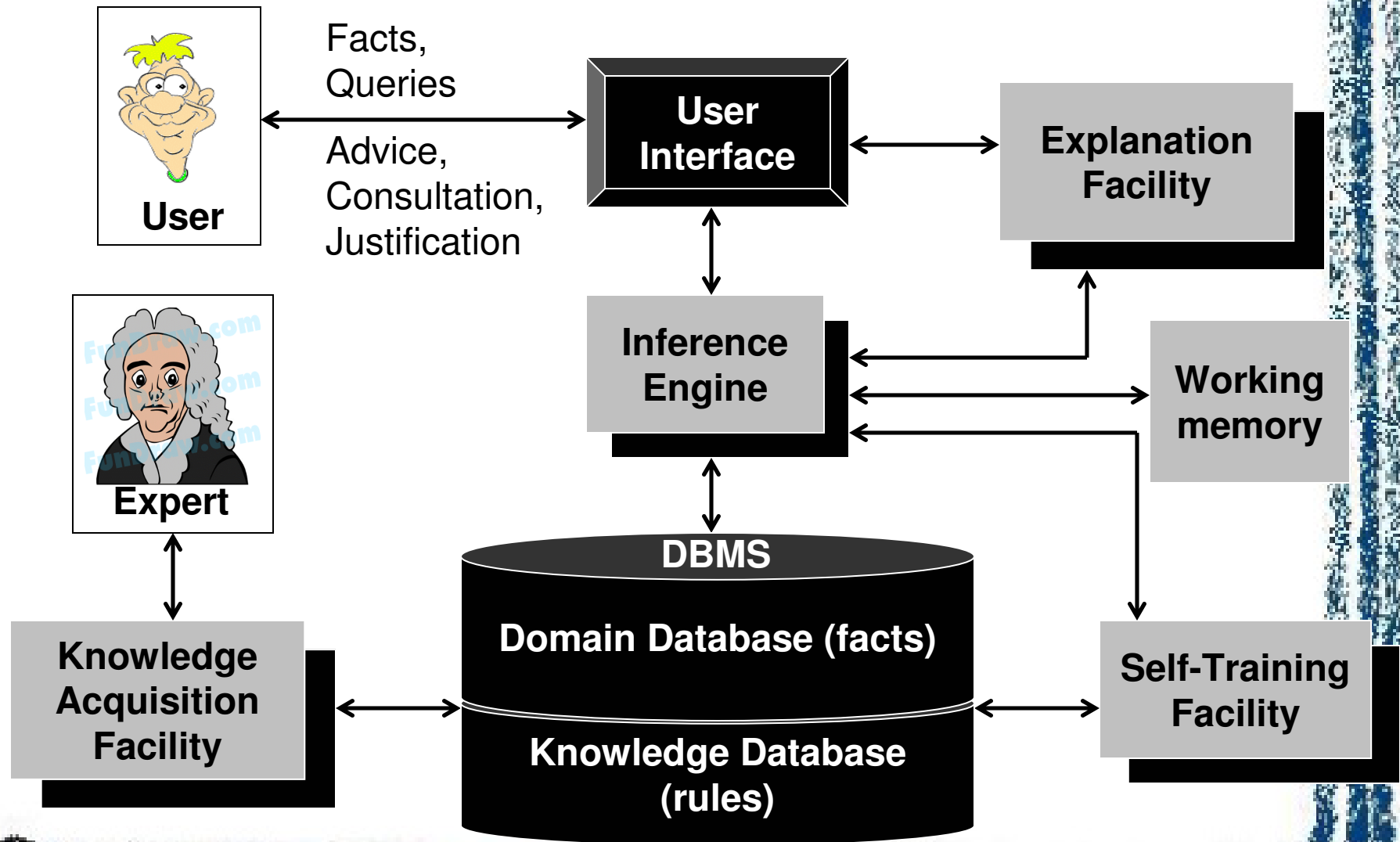


The Other Components (2)

1. Explanation facility:
 - Explain the reasoning of the system to a user
2. Knowledge acquisition facility:
 - An automatic way for the expert to enter knowledge in the system rather than by having the knowledge engineer explicitly code the knowledge
3. Self-training facility:
 - An automatic way of the system to add new facts and/or rules in the system

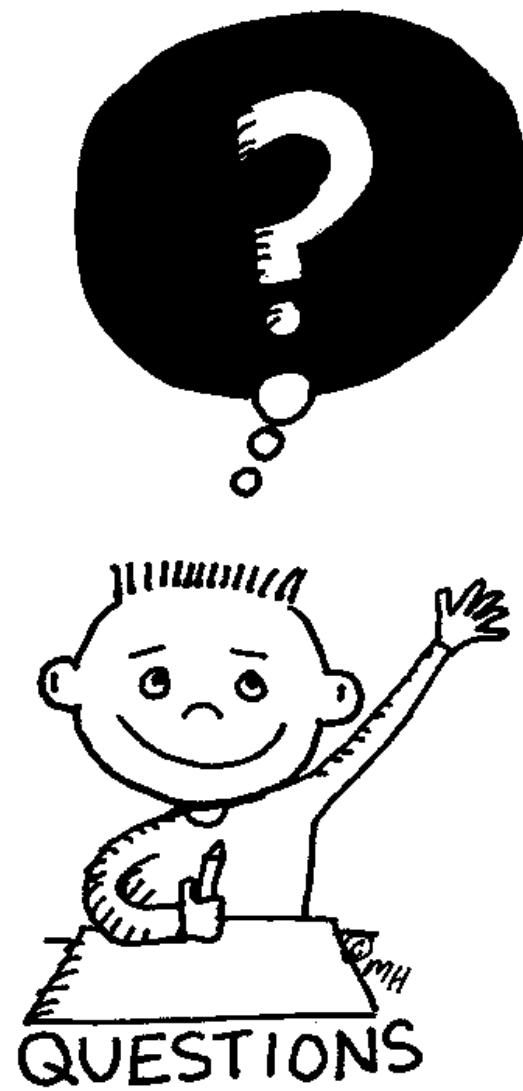
Architecture of Expert System

The complex architecture



Summary

- ES has been structured in many ways
 - The various ES architecture include different components
- There are three basic components of the ES



joko purwadi